Strand 1: Number

Sub-strand 1: Number and Numeration Systems

Lesson 2: Real Number System, Sets, and Venn Diagrams

1. Curriculum Standard and Overview

- Strand: 1 — Number
- Sub-strand: 1 — Number and Numeration Systems
- Content Standard: B9.1.1.2 — Understand the structure of the real number system and use set ideas to solve problems.
- Indicators:
- B9.1.1.2.1 — Solve problems on relationships between members of the rational number system.
- B9.1.1.2.2 — Apply the concept of sets to solve problems on relationships between members of the number system and in real-life surveys.
- Scope:
  1. The number family tree: Natural → Whole → Integers → Rational → Real.
  2. Deciding which set a number belongs to.
  3. Set language: elements, universal set, complement, union, intersection.
  4. Drawing and reading 2-set and 3-set Venn diagrams.
  5. Survey word problems (the classic "in a market of …" type).
- Mock-paper links: Paper 1 Q15 (largest set); Paper 2 Q5(d) (fishmongers Venn diagram).

2. Preamble — Numbers Live in Families

You already know counting numbers: 1, 2, 3, … But maths also needs zero, negatives, fractions, and numbers like $\sqrt{2}$ that never end.

Mathematicians group numbers into families (called sets) so we can talk about "what kind of number is this?" precisely.

Key word — SET. A set is simply a collection of well-defined objects, usually written inside curly brackets: {1, 2, 3}. Key word — ELEMENT. Each member of a set is an element. We write for "is an element of" and for "is not".

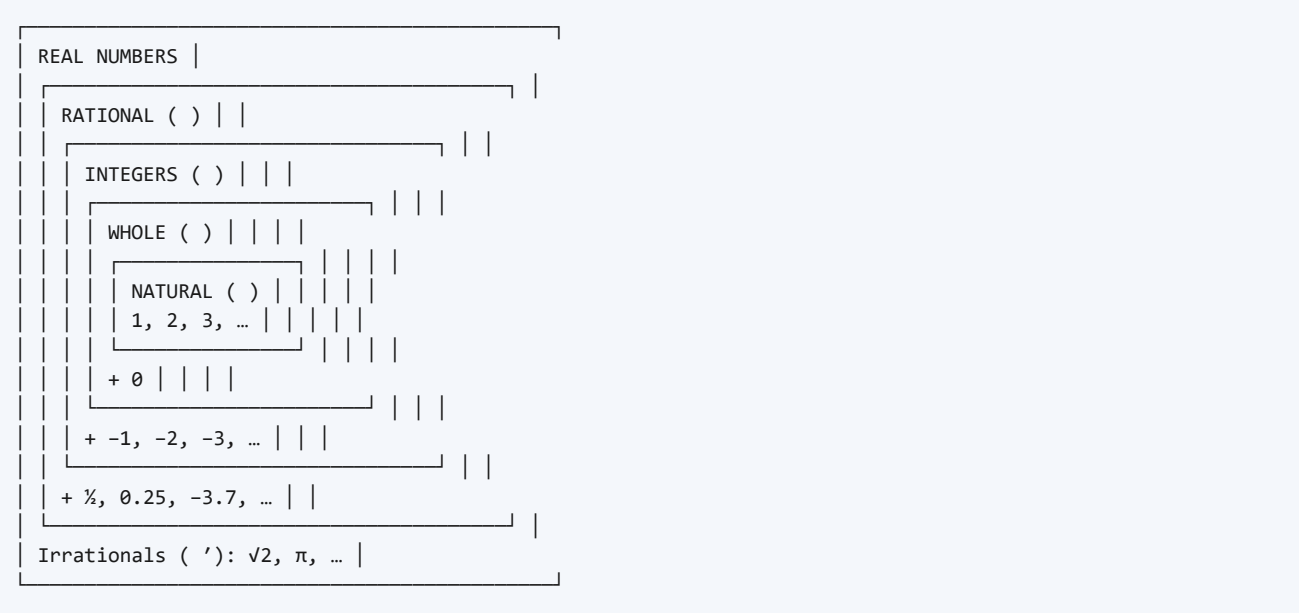
3. The Number Family Tree

3.1 The five big families

Symbol	Name	What it contains	Examples
$\mathbb{N}$	Natural (counting) numbers	1, 2, 3, 4, …	7, 100
$\mathbb{W}$	Whole numbers	Natural numbers + 0	0, 5
$\mathbb{Z}$	Integers	Whole numbers + negatives	−9, 0, 12
$\mathbb{Q}$	Rational numbers	Any number you can write as p/q (p, q integers, q ≠ 0). Includes terminating and recurring decimals	3/4, −2, 0.25
$\mathbb{Q}'$	Irrational numbers	Numbers that cannot be written as p/q; decimals go on forever without repeating	$\sqrt{2}$, π

Together, $\mathbb{Q}$ and $\mathbb{Q}'$ make the Real numbers ($\mathbb{R}$).

3.2 Nesting picture (each family is inside the one after it)



3.3 How to classify a number quickly

Ask these questions in order:

1. Is it a negative whole number? → Integer only (not whole/natural).
2. Is it a positive whole number (or zero)? → Natural and whole (and integer and rational).
3. Can it be written exactly as a fraction p/q? → Rational.
4. Does its decimal go on forever with no repeating pattern? → Irrational.

Examples:

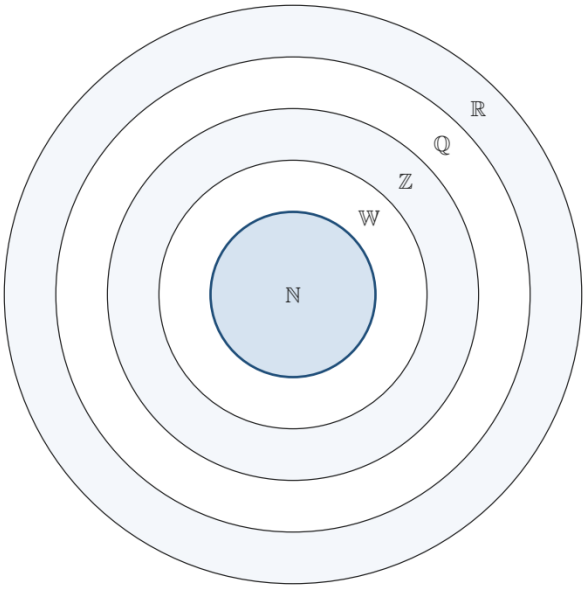
- 5 → Natural, Whole, Integer, Rational.
- 0 → Whole, Integer, Rational (not natural).
- −8 → Integer, Rational.
- 0.75 = 3/4 → Rational.
- 1.333… = 4/3 → Rational (recurring).
- $\sqrt{2} \approx 1.414213\ldots$ → Irrational.

4. Set Notation You Must Know

Symbol	Meaning	Example
$\in$	is an element of	$5 \in \{1, 3, 5\}$
$\notin$	is not an element of	$4 \notin \{1, 3, 5\}$
$\cup$	union (everything in either set)	$\{1,2\} \cup \{2,3\} = \{1,2,3\}$
$\cap$	intersection (only what is in both)	$\{1,2\} \cap \{2,3\} = \{2\}$
A' or A^c	complement of A (not in A, but in U)	If U = {1..6}, A = {1,2}, $A' = \{3,4,5,6\}$
U or ξ	universal set (everything we are talking about)	
n(A)	number of elements in A (the count)	$n(\{2,4,6\}) = 3$
$\subset$	is a subset of	$\mathbb{N} \subset \mathbb{Z}$
$\emptyset$ or {}	empty set	

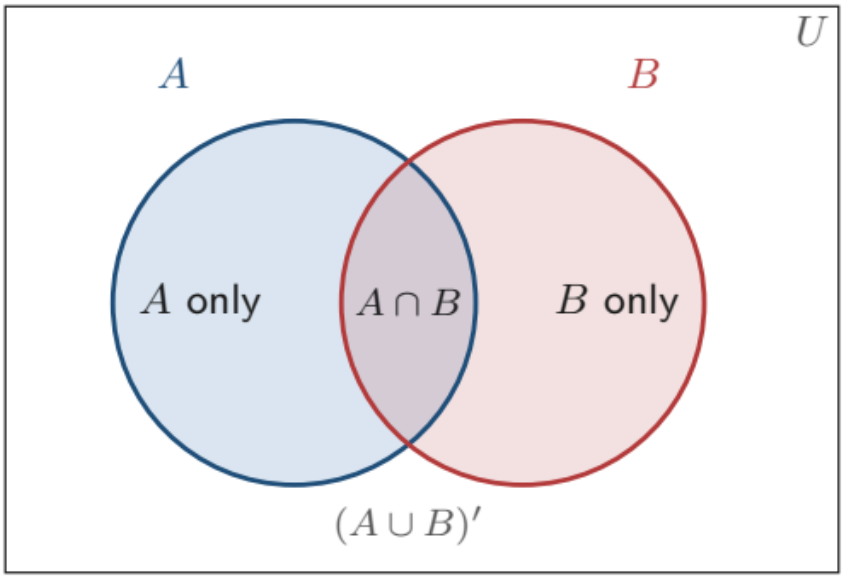
5. Venn Diagrams

5.1 What a Venn diagram does



A Venn diagram is a picture of sets. A rectangle is the universal set. Each circle is one set. The overlapping region is the intersection.

5.2 Two-set Venn diagram layout




```
┌──────────────────────────────────┐
│ Universal set U                  │
│  ┌──────────┐ ┌──────────┐       │
│  │ only A   │ A │ only B   │     │
│  │          │ ∩ │          │     │
│  │          │ B │          │     │
│  └──────────┘ └──────────┘       │
│ outside both = (A B)'            │
└──────────────────────────────────┘
```

5.3 The golden counting rule (2 sets)

For two sets A and B inside a universal set of size n(U):

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

We subtract the overlap once because adding n(A) and n(B) counts the overlap twice.

Rearranged:

$$n(A \text{ only}) = n(A) - n(A \cap B)$$
$$n(B \text{ only}) = n(B) - n(A \cap B)$$
$$n(\text{neither}) = n(U) - n(A \cup B)$$

Always fill a Venn from the middle (the overlap) outwards.

6. Twenty Fully Worked Examples

Example 1 — Classify 7

7 is a counting number → Natural, Whole, Integer, Rational.

Example 2 — Classify 0

0 is not a counting number → Whole, Integer, Rational (not natural).

Example 3 — Classify −12

Integer, Rational (also rational because −12 = −12/1).

Example 4 — Is 0.25 rational?

0.25 = ¼, so yes — Rational.

Example 5 — Is $\sqrt{9}$ rational?

$\sqrt{(9)}$ = 3, which is a natural number → Rational. (Careful: the square root symbol does not automatically mean irrational.)

Example 6 — Is $\sqrt{7}$ rational?

7 is not a perfect square, so $\sqrt{7}$ = 2.6457… (no repeating block) → Irrational.

Example 7 — Largest of these sets

Which is the largest set: {whole numbers}, {composite numbers}, {natural numbers}, {integers}?

- Natural = {1,2,3,…}
- Whole = {0,1,2,3,…}
- Integers = {…,−2,−1,0,1,2,…}
- Composite = {4,6,8,9,10,…} (smallest family here)

Answer: {integers} (Mock 2026 P1 Q15 — option D).

Example 8 — Smallest of those sets

From the list above, the composite numbers form the smallest set.

Example 9 — Simple union

If A = {1, 2, 3} and B = {3, 4, 5}, find A ∪ B. A ∪ B = {1, 2, 3, 4, 5}.

Example 10 — Simple intersection

Using the same A and B: A ∩ B = {3}.

Example 11 — Complement

U = {1,2,3,4,5,6}, A = {1,2}. Find A′. A′ = {3, 4, 5, 6}.

Example 12 — Two-set counts

In a class of 40, 22 play football, 18 play hockey, 9 play both. How many play neither?

1. n(F ∪ H) = 22 + 18 - 9 = 31.
2. Neither = 40 - 31 = 9.

Answer: 9 students.

Example 13 — Fill the Venn from Example 12

Football only = 22 - 9 = 13. Hockey only = 18 - 9 = 9. Both = 9. Neither = 9. Check: 13 + 9 + 9 + 9 = 40 — correct.

Example 14 — Mock Paper 2 Q5(d): fishmongers

In a market of 36 fishmongers, 28 sell mackerel (M), 18 sell tilapia (T), 13 sell both.

(i) Venn diagram:

- Both: 13
- Mackerel only: 28 - 13 = 15
- Tilapia only: 18 - 13 = 5
- Outside both: 36 - (15+13+5) = 3

(ii) Not selling any of the two: 3.

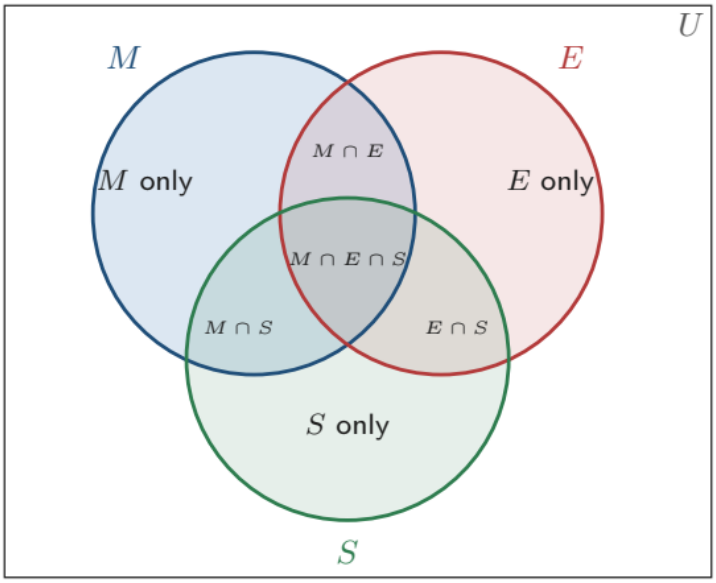


Example 15 — Finding n(A) from the picture

From Example 14, how many sell mackerel? Mackerel only + both = 15 + 13 = 28 (matches the question).

Example 16 — Three-set introduction

If we add a third set C (say, those who sell crabs), we draw a third circle. The centre region is A ∩ B ∩ C (all three). We still fill the innermost regions first.



Example 17 — Subset

True or false: $\mathbb{N} \subset \mathbb{Z}$? True — every natural number is an integer.

Example 18 — True/False on membership

Is $-3 \in$ {whole numbers}? No → False (whole numbers start at 0).

Example 19 — Word → set

"The set of even numbers between 1 and 11." = {2, 4, 6, 8, 10}.

Example 20 — Survey with percentages

In a survey of 200 people, 120 like tea, 90 like coffee, and 40 like both. How many like neither?

n(T ∪ C) = 120 + 90 - 40 = 170. Neither = 200 - 170 = 30.

Answer: 30 people.

7. Comprehensive Exercise Sets (40 Questions)

All authentic BECE/Mock questions for this topic are included in full. Generated questions top up to the 40-quota. Spiral review ≈ 15%.

Exercise 1 — Identifying number families

1. [Mock 2026 P1 Q15] Which of the following sets is the largest?
   A. {whole numbers} B. {composite numbers} C. {natural numbers} D. {integers}
2. Which of these sets is the smallest?
   A. {whole numbers} B. {composite numbers} C. {natural numbers} D. {integers}
3. Classify 5 under every family that fits ($\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$).
4. Classify -3 under every family that fits.
5. Classify 0 under every family that fits.

Exercise 2 — Rational vs irrational

1. Is 0.25 rational? Give a reason.
2. Is $\sqrt{9}$ rational? Give a reason.
3. Is $\sqrt{7}$ rational or irrational?
4. Write 0.333… as a fraction; hence show it is rational.

5. Spiral review: Write 47 300 in standard form (Lesson 1 revision).

Exercise 3 — Set notation

1. If A = {1, 2, 3}, state whether 2 ∈ A.
2. If A = {1, 2, 3}, state whether 4 ∈ A.
3. U = {1,2,3,4,5,6}, A = {1,2}. Find A′.
4. Let X = {2,4,6}, Y = {4,6,8}. Find X ∩ Y.
5. Let X = {2,4,6}, Y = {4,6,8}. Find X ∪ Y.

Exercise 4 — Counting elements

1. Find n({3, 5, 7, 9}).
2. Find n(∅).
3. If n(A) = 7 and n(B) = 5 and A ∩ B = {2}, find n(A ∪ B).
4. Spiral review: Evaluate 36 ÷ (4 + 2) × 5 (BODMAS).
5. Spiral review: Round 0.006 845 to 2 s.f.

Exercise 5 — Two-set Venn diagrams (counts)

1. In a class of 40, 22 play football, 18 play hockey, 9 play both. How many play neither?
2. For the same class, how many play football only?
3. For the same class, how many play hockey only?
4. In a group of 50, 30 drink tea, 25 drink coffee, 10 drink both. How many drink neither?
5. In a group of 50, 30 drink tea, 25 drink coffee, 10 drink both. How many drink tea only?

Exercise 6 — Survey problems (the mock style)

1. [Mock 2026 P2 Q5(d)] In a market of 36 fishmongers, 28 sell mackerel, 18 sell tilapia and 13 of them sell both.

(i) Represent this information on a Venn diagram. (ii) How many in the market are not selling any of the two fishes?

2. In a survey of 200 people, 120 like tea, 90 like coffee, and 40 like both. How many like neither?
3. In a school of 180 pupils, 100 study French, 80 study Twi, and 30 study both. How many study neither?
4. From Question 3, how many study French only?
5. Spiral review: Convert 31 days into minutes and write the answer in standard form.

Exercise 7 — Subsets and true/false

1. True or false: ℕ ⊂ ℤ?
2. True or false: −3 ∈ {whole numbers}?
3. Write "the set of even numbers between 1 and 11" in curly-bracket notation.
4. List all elements of {x : x is a factor of 12}.
5. Spiral review: Write 5 040 007 in words.

Exercise 8 — Mixed and challenge

1. If U = {1,2,3,4,5,6,7,8,9,10}, A = {2,4,6,8,10}, B = {3,6,9}, find

(i) A ∩ B (ii) A ∪ B (iii) A′.

2. In a class of 35, every pupil studies at least one of Maths (M) and Science (S). 20 study M, 18 study S, and 8 study both. How many study Maths only?
3. A Venn diagram shows n(U) = 60, n(A) = 25, n(B) = 30, n(A ∪ B) = 45. Find n(A ∩ B).
4. Spiral review: Arrange in ascending order: 3 050 082; 3 005 820; 3 500 082; 3 050 802.
5. Spiral review: Find the value of the digit 6 in 9 463 285.

8. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. Integers contain negatives, zero, and all whole numbers — the biggest family listed. Answer: D.
2. Composite numbers (4, 6, 8, 9, …) are the most restricted. Answer: B.
3. 5 is a counting number, so $5 \in \mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}$.
4. -3 is not counting/whole $\rightarrow -3 \in \mathbb{Z}, \mathbb{Q}$ only.
5. 0 is whole but not natural $\rightarrow 0 \in \mathbb{W}, \mathbb{Z}, \mathbb{Q}$.

Solutions to Exercise 2

1. 0.25 = ¼ (p/q form) → Rational.
2. $\sqrt{(9)} = 3 \in \mathbb{N} \rightarrow$ Rational.
3. $\sqrt{7} \approx 2.645751\ldots$ with no repeating block → Irrational.
4. Let x = 0.333ldots; then 10x = 3.333ldots; subtract: $9x = 3 \rightarrow x = \frac{1}{3} \rightarrow$ Rational.
5. 47 300 = 4.73 × 10^(4).

Solutions to Exercise 3

1. 2 is listed in A → True.
2. 4 is not listed in A → False.
3. $A' = \{3, 4, 5, 6\}$.
4. $X \cap Y = \{4, 6\}$.
5. $X \cup Y = \{2, 4, 6, 8\}$.

Solutions to Exercise 4

1. Count the elements: 4.
2. The empty set has 0 elements.
3. $n(A \cup B) = 7 + 5 - 1 = 11$.
4. 36 ÷ 6 = 6; 6 × 5 = 30 → 30.
5. 0.006 845 → 2 s.f. → 0.0068.

Solutions to Exercise 5

1. $n(F \cup H) = 22 + 18 - 9 = 31$; neither = 40 - 31 = 9.
2. Football only = 22 - 9 = 13.
3. Hockey only = 18 - 9 = 9.
4. $n(T \cup C) = 30 + 25 - 10 = 45$; neither = 50 - 45 = 5.
5. Tea only = 30 - 10 = 20.

Solutions to Exercise 6

1. (i) Overlap 13; M only = 28-13 = 15; T only = 18-13 = 5. (ii) Outside = 36 - (15+13+5) = 3.
2. 120 + 90 - 40 = 170; neither = 200 - 170 = 30.
3. 100 + 80 - 30 = 150; neither = 180 - 150 = 30.
4. French only = 100 - 30 = 70.
5. 31 × 1440 = 44 640 = 4.464 × 10^(4) minutes.

Solutions to Exercise 7

1. Every natural number is an integer → True.
2. Whole numbers are 0, 1, 2, … → False.
3. {2, 4, 6, 8, 10}.
4. Factors of 12: 1, 2, 3, 4, 6, 12 → {1, 2, 3, 4, 6, 12}.
5. 5 040 007 = five million, forty thousand, and seven.

Solutions to Exercise 8

1. (i) $A \cap B = \{6\}$ (ii) $A \cup B = \{2, 3, 4, 6, 8, 9, 10\}$ (iii) $A' = \{1, 3, 5, 7, 9\}$
2. Maths only = 20 - 8 = 12.
3. $n(A \cap B) = 25 + 30 - 45 = 10$.
4. 3 005 820 < 3 050 082 < 3 050 802 < 3 500 082.
5. Digit 6 is in the ten-thousands place → 60 000.

9. Authentic BECE & Mock Questions Used

Bank / source	Question	Answer
Mock 2026 P1 Q15	Largest set?	D — {integers}
Mock 2026 P2 Q5(d)	Fishmongers Venn: 36 total, 28 M, 18 T, 13 both	Neither = 3

10. Lesson Summary

Skill	Rule
Classify numbers	Ask: integer? whole? fraction? non-repeating decimal?
Union	Everything in either set
Intersection	Only the overlap
Complement	Inside U but outside the set
Two-set count	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
Venn strategy	Fill from the middle outwards

Lesson prepared under `Grade9-Maths/Rules.md` — plain English, British notation, full workings required.